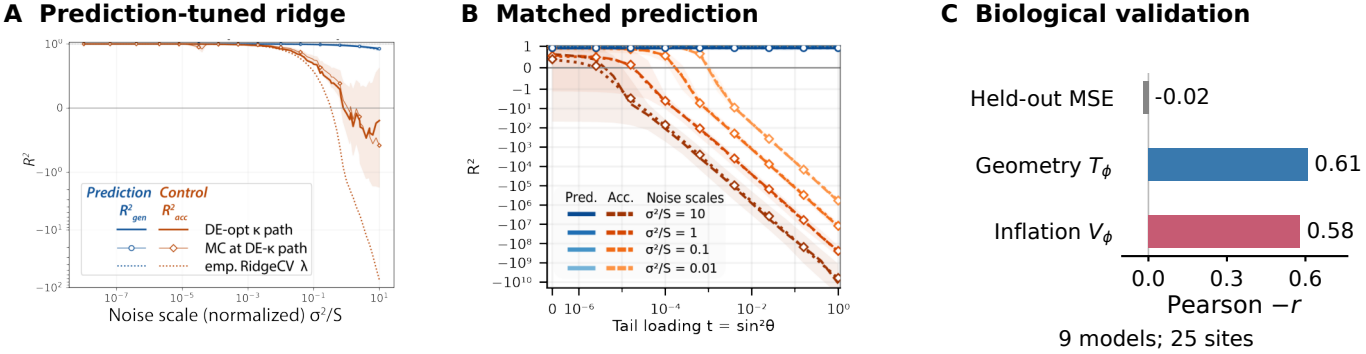

A Prediction-tuned ridge
B Matched prediction
C Biological validation
Prediction
Control
DE-opt κ path
MC at DE-κ path
emp. RidgeCV λ
Noise scale (normalized) σ²/S
Pred.
Acc.
Noise scales
σ²/S = 10
σ²/S = 1
σ²/S = 0.1
σ²/S = 0.01
Tail loading t = sin²θ
Held-out MSE
-0.02
Geometry T_ϕ
0.61
Inflation V_ϕ
0.58
Pearson −r
9 models; 25 sites